

Additionneur 2 bits en SystemC

half_adder.h

```
// File : half_adder.h  
#include "systemc.h"
```

Nom du module

```
SC_MODULE(half_adder)
```

```
{
```

```
    sc_in<bool> a,b;
```

```
    sc_out<bool> sum,carry;
```

Données membres, l'interface

```
    void prc_half_adder();
```

```
    SC_CTOR(half_adder)
```

```
{
```

```
        SC_METHOD(prc_half_adder);
```

```
        sensitive << a << b;
```

```
}
```

```
};
```

Constructeur

half_adder.h

SC_METHOD (1)

```
// File : half_adder.h
#include "systemc.h"

SC_MODULE(half_adder)
{
    sc_in<bool> a,b;
    sc_out<bool> sum,carry;

    void prc_half_adder();

    SC_CTOR(half_adder)
    {
        SC_METHOD(prc_half_adder);
        sensitive << a << b;
    }
};
```

Déclare un process sans
mémoire appelé prc_half_adder()

Ne peut utiliser les wait

Sensibilité « statique » sur a et b

SC_METHOD (2)

```
// File : half_adder.cpp
#include "half_adder.h"

void half_adder::prc_half_adder()
{
    sum = a ^ b;
    carry = a & b;
}
```

Description de la hiérarchie

```
// File : full_adder.h
#include "half_adder.h"

SC_MODULE(full_adder)
{
    sc_in<bool> a,b,carry_in;
    sc_out<bool> sum,carry_out;

    sc_signal<bool> c1,s1,c2;
    void prc_or();

    half_adder *ha1_ptr,*ha2_ptr;

    SC_CTOR(full_adder)
    {
        ha1_ptr=new half_adder("ha1");
        // Named association:
        ha1_ptr->a(a);
        ha1_ptr->b(b);
        ha1_ptr->sum(s1);
        ha1_ptr->carry(c1);

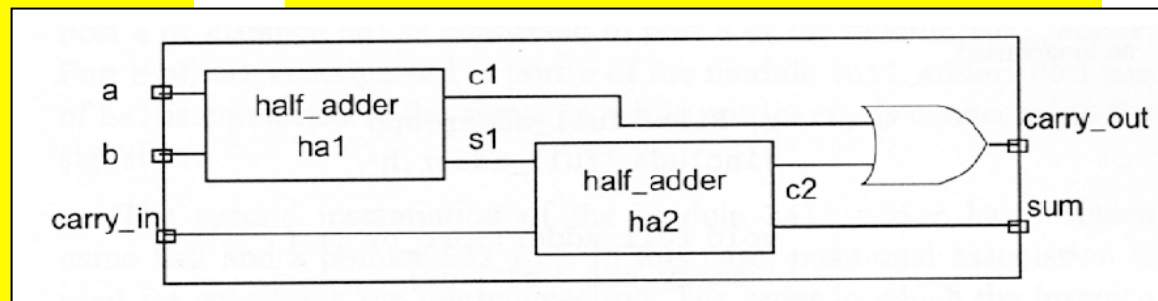
        ha2_ptr=new half_adder("ha2");
        // Positional association:
        (*ha2_ptr)(s1,carry_in,sum,c2);

        SC_METHOD(prc_or);
        sensitive << c1 << c2;
    }

    // A destructor
    ~full_adder()
    {
        delete ha1_ptr;
        delete ha2_ptr;
    }
};
```

```
// File : full_adder.cpp
#include "full_adder.h"

void full_adder::prc_or()
{
    carry_out = c1 | c2;
}
```



driver.h, driver.cpp

```
// File : driver.h
#include "systemc.h"

SC_MODULE(driver)
{
    sc_out<bool> d_a,d_b,d_cin;

    void prc_driver();

    SC_CTOR(driver)
    {
        SC_THREAD(prc_driver);
    }
};
```

```
// File : driver.cpp
#include "driver.h"

void driver::prc_driver()
{
    sc_uint<3> pattern;
    pattern=0;

    while (1)
    {
        d_a=pattern[0];
        d_b=pattern[1];
        d_cin=pattern[2];
        wait(5,SC_NS);
        pattern++;
    }
}
```

monitor.h, monitor.cpp

```
// File : monitor.h
#include "systemc.h"

SC_MODULE(monitor)
{
    sc_in<bool>
    m_a,m_b,m_cin,m_sum,m_cout;

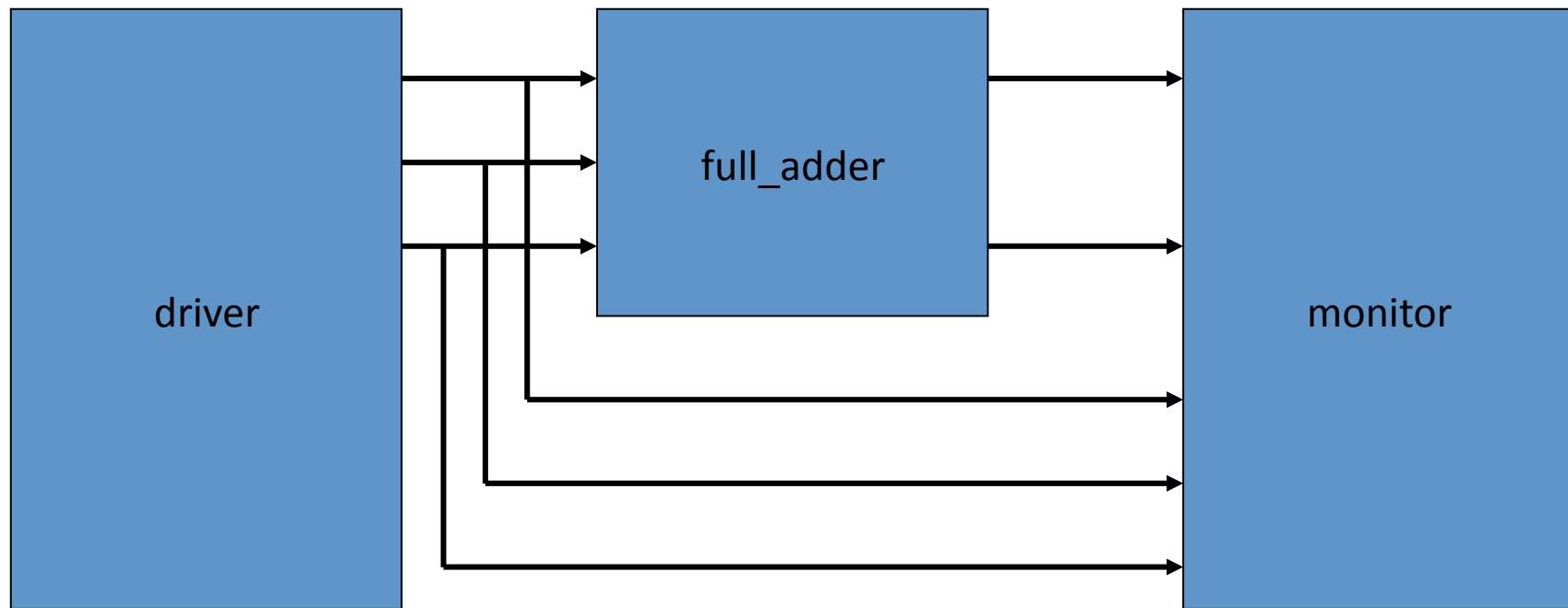
    void prc_monitor();

    SC_CTOR(monitor)
    {
        SC_METHOD(prc_monitor);
        sensitive << m_a
                << m_b
                << m_cin
                << m_sum
                << m_cout;
    }
};
```

```
// File : monitor.cpp
#include "monitor.h"

void monitor::prc_monitor()
{
    cout << "At time " <<
    sc_time_stamp() << " :: ";
    cout << "(a, b, carry_in): ";
    cout << m_a
        << m_b << m_cin;
    cout << " (sum, carry_out): "
        << m_sum
        << m_cout << endl;
}
```

Schématique du toplevel



Le toplevel

```
// File : full_adder_main.cpp
#include "driver.h"
#include "monitor.h"
#include "full_adder.h"

int sc_main(int argc, char *argv[])
{
    sc_signal<bool> t_a, t_b, t_cin, t_sum, t_cout;

    full_adder f1("FullAdderWithHalfAdder");
    // Connect using positional association:
    f1 << t_a << t_b << t_cin << t_sum << t_cout;

    driver d1("GenerateWaveforms");
    // Connect using named association:
    d1.d_a(t_a);
    d1.d_b(t_b);
    d1.d_cin(t_cin);

    monitor m1("MonitorWaveforms");
    m1 << t_a << t_b << t_cin << t_sum << t_cout;

    sc_start(100, SC_NS);

    return(0);
}
```

← A éviter

← A utiliser

```
[fpecheux@bip]/users/chaos/SYSC/TPl/adder% ./run.x
```

```
SystemC 2.2.0 --- Sep 24 2009 13:11:09  
Copyright (c) 1996-2006 by all Contributors  
ALL RIGHTS RESERVED
```

```
Info: (I804) /IEEE_Std_1666/deprecated: positional binding using << or , is deprecated, use () instead.
```

```
At time 0 s::(a, b, carry_in): 000 (sum, carry_out): 00  
At time 5 ns::(a, b, carry_in): 100 (sum, carry_out): 00  
At time 5 ns::(a, b, carry_in): 100 (sum, carry_out): 10  
At time 10 ns::(a, b, carry_in): 010 (sum, carry_out): 10  
At time 15 ns::(a, b, carry_in): 110 (sum, carry_out): 10  
At time 15 ns::(a, b, carry_in): 110 (sum, carry_out): 01  
At time 20 ns::(a, b, carry_in): 001 (sum, carry_out): 01  
At time 20 ns::(a, b, carry_in): 001 (sum, carry_out): 11  
At time 20 ns::(a, b, carry_in): 001 (sum, carry_out): 10  
At time 25 ns::(a, b, carry_in): 101 (sum, carry_out): 10  
At time 25 ns::(a, b, carry_in): 101 (sum, carry_out): 00  
At time 25 ns::(a, b, carry_in): 101 (sum, carry_out): 01  
At time 30 ns::(a, b, carry_in): 011 (sum, carry_out): 01  
At time 35 ns::(a, b, carry_in): 111 (sum, carry_out): 01  
At time 35 ns::(a, b, carry_in): 111 (sum, carry_out): 11  
At time 40 ns::(a, b, carry_in): 000 (sum, carry_out): 11  
At time 40 ns::(a, b, carry_in): 000 (sum, carry_out): 01  
At time 40 ns::(a, b, carry_in): 000 (sum, carry_out): 00  
At time 45 ns::(a, b, carry_in): 100 (sum, carry_out): 00  
At time 45 ns::(a, b, carry_in): 100 (sum, carry_out): 10  
At time 50 ns::(a, b, carry_in): 010 (sum, carry_out): 10  
At time 55 ns::(a, b, carry_in): 110 (sum, carry_out): 10  
At time 55 ns::(a, b, carry_in): 110 (sum, carry_out): 01  
At time 60 ns::(a, b, carry_in): 001 (sum, carry_out): 01  
At time 60 ns::(a, b, carry_in): 001 (sum, carry_out): 11  
At time 60 ns::(a, b, carry_in): 001 (sum, carry_out): 10  
At time 65 ns::(a, b, carry_in): 101 (sum, carry_out): 10  
At time 65 ns::(a, b, carry_in): 101 (sum, carry_out): 00  
At time 65 ns::(a, b, carry_in): 101 (sum, carry_out): 01  
At time 70 ns::(a, b, carry_in): 011 (sum, carry_out): 01  
At time 75 ns::(a, b, carry_in): 111 (sum, carry_out): 01  
At time 75 ns::(a, b, carry_in): 111 (sum, carry_out): 11  
At time 80 ns::(a, b, carry_in): 000 (sum, carry_out): 11  
At time 80 ns::(a, b, carry_in): 000 (sum, carry_out): 01  
At time 80 ns::(a, b, carry_in): 000 (sum, carry_out): 00  
At time 85 ns::(a, b, carry_in): 100 (sum, carry_out): 00  
At time 85 ns::(a, b, carry_in): 100 (sum, carry_out): 10  
At time 90 ns::(a, b, carry_in): 010 (sum, carry_out): 10  
At time 95 ns::(a, b, carry_in): 110 (sum, carry_out): 10  
At time 95 ns::(a, b, carry_in): 110 (sum, carry_out): 01
```

```
Info: (I804) /IEEE_Std_1666/deprecated: You can turn off warnings about  
IEEE 1666 deprecated features by placing this method call as the  
first statement in your sc_main() function:
```

```
sc_report_handler::set_actions("/IEEE_Std_1666/deprecated", SC_DO_NOTHING);
```

Trace de simulation

Pour ajouter des traces (.vcd)

```
sc_initialize();

//      (1) Ouvrir le fichier de trace, au format VCD

sc_trace_file *tfile=sc_create_vcd_trace_file("results");

//      (2) Rajouter les signaux que l'on souhaite tracer
//      Le premier argument est un pointeur de fichier
//      Le deuxième est le nom du signal.
//      Le troisième est le nom avec lequel le signal va être
//      affiché par gtkwave. Les espaces dans le nom sont
//      interdits

sc_trace(tfile,t_a,"t_a") ;
sc_trace(tfile,t_b,"t_b") ;
sc_trace(tfile,t_cin,"t_cin") ;
sc_trace(tfile,t_sum,"t_sum") ;
sc_trace(tfile,t_cout,"t_cout") ;

...      (3) Lancer la simulation : sc_start() ou sc_cycle()

//      (4) Fermer le fichier de traces

sc_close_vcd_trace_file(tfile);

return(0);
```